

CANDIDATE KNOWLEDGE BASE

Build Challenge Reference Guide · SuperCharge SG Business & Technology Context
Tech Stack Tips · Architecture Guides · Singapore Energy Domain Knowledge
AGV Hardware Reference · RAG Source Material for Companion Chatbot

This Knowledge Base is your companion resource for the SuperCharge SG build challenge. It contains everything you need to understand the business context, make the right technical decisions, and build with confidence. It has also been ingested into the candidate support chatbot — so you can ask the bot any question and it will draw its answers from this document. Read it once end-to-end before you start building. Come back to specific sections as needed.

Section	Title	Covers
A	SuperCharge SG — Company, Products & Business Model	Who we are, what we sell, how we earn revenue
B	Singapore Energy Regulatory Context	TR25, OCPP, ECIS, SP Group, LTA — the rules you must know
C	Challenge 1: Chatbot — Architecture & Tips	Telegram, WhatsApp API, RAG pipeline, deployment
D	Challenge 2: Smart Energy Dashboard — Architecture & Tips	Data pipeline, ML anomaly, multi-tenant, ECIS calc
E	Challenge 3: AGV Brain — Design Reference	TADA B1900, PWM intercept, navigation stacks, BOM guide
F	Evaluation Criteria & Scoring Detail	What earns full marks in each challenge
G	Singapore Market & Domain Facts	Quick-reference facts for grounding your chatbot KB

SECTION A — SUPERCHARGE SG: COMPANY, PRODUCTS & BUSINESS MODEL

A1. Who We Are

SuperCharge SG (supercharge.sg) is a Singapore-based clean energy startup and LTA-licensed Electric Vehicle Charging Operator (EVCO). We deliver end-to-end EV charging and solar PV installation services to homeowners, businesses, and industrial clients across Singapore. Our mission is to make clean energy infrastructure accessible, affordable, and intelligently managed. We operate under LTA's EVCO licence framework and comply with TR25 — Singapore's technical reference standard for EV charging installations.

We are an impact-driven startup. We partner with ITE College East to support smart energy training programmes, deploy chargers for premium residential developments (The Coast @ Sentosa Cove), automotive showrooms (Cupra), and commercial properties. Our team includes Licensed Electrical Workers (LEWs) certified for TR25 compliance and Equipment Specialists who handle site assessment, installation, and LTA 6-monthly inspections.

A2. Products & Services

EV Charging

Product/Service	Description	Key Feature	Revenue Model
TR25-Compliant EV Charger Supply & Install	Residential and commercial EV charger hardware supply and certified installation	LTA-compliant, LEW-certified, 6-monthly inspection support	One-time install fee + maintenance contract
OCPP/OCPI Backend Platform	Smart charger management: real-time status, alerts, remote control, session logs	OCPP 1.6/2.0 compliant, multi-site scalable	SaaS subscription per charger/site
Smart Alerts & Notifications	Proactive charger fault detection and push notification to operators	AI-powered uptime management	Included in SaaS tier
LTA 6-Monthly Inspection Service	Mandatory compliance inspection and certification by LEW	Required by LTA for all EVCO operators	Per-inspection fee
Multi-site Fleet Management	Centralised control of chargers across multiple locations	OCPI-enabled for network interoperability	Enterprise SaaS licence

Solar PV Installation

Product/Service	Description	Key Claim
All-Black Solar Panel Systems	High-performance sleek solar panel installation for landed homes and commercial rooftops	Up to 30% reduction in electricity bills
PV Optimiser Integration	Panel-level power optimisation to reduce mismatch losses and soiling impact	Increases energy yield from shaded or complex rooftops
SP Group Approvals & ECIS Setup	Full paperwork management including SP Group export application and ECIS credit setup	Hassle-free — client just sits back
25-Year System Warranty	Long-term panel and inverter performance warranties	Tens of thousands SGD savings over system lifetime
Solar + EV Bundle	Combined solar and EV charger installation with integrated monitoring	Single provider, optimised for self-consumption charging

A3. Customer Profiles & Common Questions

Customer Type	Common Questions / Needs	What They Care About
Homeowner (Landed)	How much does a solar system cost? How long to install? Can I export to grid?	Savings-focused, wants simple payback period numbers
Homeowner (HDB/Condo)	Can I install EV charger in my condo carpark? Who approves this?	Needs LTA/MCST process guidance
Business / Commercial	Multi-site charger management, invoice reports, uptime SLAs	Operations and cost-efficiency focused

Industrial / Facility Mgr	ESG reporting, energy monitoring, TR25 compliance certificates	Compliance and data reporting focused
Automotive / Showroom	Fast charger install, customer-facing charging, brand-matched hardware	Brand image and client experience
Education Institution	Training setups, grant eligibility, smart energy demonstrations	Grants (ECCG) and educational context

A4. Pricing Reference (Public Estimates)

These are indicative ranges for your chatbot knowledge base. Direct prospects to contact SuperCharge for exact quotes.

Item	Indicative Cost / Rate (SGD)
EV Charger (7kW AC, Type 2)	From SGD 1,200–2,500 installed (depending on cable run length and DB work required)
EV Charger (22kW AC, 3-phase)	From SGD 2,500–4,500 installed
DC Fast Charger (50kW+)	From SGD 15,000–40,000+ depending on infrastructure requirements
Solar System (3–5kWp, landed home)	From SGD 8,000–14,000 after subsidies/grants (highly variable)
Solar System (10–30kWp, commercial)	From SGD 25,000–80,000+ (project-dependent)
EV charging cost to end user	Approximately SGD 0.50/kWh average (varies by operator and time of day)
ECIS solar export credit	SGD 0.218/kWh (Enhanced Central Intermediary Scheme baseline — verify current rate with SP Group)
LTA 6-monthly inspection fee	From SGD 150–300 per inspection visit (varies by site complexity)

SECTION B — SINGAPORE ENERGY REGULATORY CONTEXT

B1. TR25 — Technical Reference for EV Charging Installations

What it is: TR25 is Singapore's national technical standard for EV charging installations. Published by Enterprise Singapore (ESG), it defines electrical safety, equipment specifications, installation requirements, and testing procedures for all EV charging points. Every EV charger installed by an EVCO like SuperCharge must comply with TR25.

- All EV chargers must be installed by a **Licensed Electrical Worker (LEW)** certified for EV work
- Chargers must be TR25-compliant hardware — not all chargers sold in Singapore are compliant
- Mandatory LTA 6-monthly inspection of all EVCO-operated charging points
- Electrical supply must meet cable sizing, protective device ratings, and earthing requirements per TR25
- Metering and billing systems must be accurate and tamper-resistant
- OCPP compliance required for smart charging interoperability
- Type 2 (IEC 62196) is the standard connector for AC charging in Singapore
- CCS2 (Combined Charging System) for DC fast charging

■ Chatbot tip

When a customer asks about 'compliance' or 'safety', your bot should reference TR25 and explain that SuperCharge's LEWs handle all certification. Never give specific electrical engineering advice — route to human agent.

B2. OCPP & OCPI — Smart Charging Protocols

OCPP (Open Charge Point Protocol): the communication protocol between EV chargers and the backend management system. SuperCharge uses OCPP 1.6 and 2.0. It enables remote start/stop, status monitoring, firmware updates, and smart charging (load management). OCPP is open-source and vendor-neutral.

OCPI (Open Charge Point Interface): the protocol for roaming between different charging networks. Enables a driver registered with one network to charge on another. SuperCharge's backend is OCPI-compliant for network interoperability.

■ Dashboard tip

For Challenge 2 (Smart Energy Dashboard), your simulated charger telemetry should mimic OCPP status messages: Available, Preparing, Charging, SuspendedEV, Finishing, Faulted, Unavailable. Use these as your charger state enum in the data pipeline.

B3. ECIS — Enhanced Central Intermediary Scheme (Solar Export Credits)

ECIS is SP Group's scheme that allows solar panel owners to sell excess electricity back to the grid and receive credits on their electricity bill. Key facts:

- Export rate: approximately **SGD 0.218/kWh** (baseline — check SP Group website for current rate as it changes quarterly)
- Credits appear on the customer's SP electricity bill as a deduction
- Requires SP Group approval before solar system is connected to grid
- SuperCharge handles all SP Group paperwork as part of the solar installation service
- Maximum export capacity is limited by the contracted supply capacity of the premises
- Residential systems typically export 20–40% of total generation depending on self-consumption
- A 5kWp system in Singapore generates approximately 550–650 kWh/month (varies with shade, orientation)
- Annual ECIS credit for a 5kWp system: approximately SGD 1,400–1,700/year at current export rate

B4. Singapore Solar Irradiance Data

For accurate solar yield calculations in Challenge 2 (Smart Energy Dashboard) and the Solar ROI model, use real Singapore irradiance data. The most reliable free source is the NASA POWER API.

```
# NASA POWER API – Singapore irradiance data # Coordinates: Singapore approx lat=1.3521, lon=103.8198 #
ALLSKY_SFC_SW_DWN = Surface irradiance (kWh/m2/day) import requests url =
'https://power.larc.nasa.gov/api/temporal/monthly/point' params = { 'parameters': 'ALLSKY_SFC_SW_DWN', 'community':
'RE', 'longitude': 103.8198, 'latitude': 1.3521, 'start': '2020', 'end': '2024', 'format': 'JSON' } r =
requests.get(url, params=params) data = r.json() # Monthly avg irradiance kWh/m2/day # Singapore average: ~4.5-5.2
kWh/m2/day (varies by month) # Performance Ratio (PR) for typical SG install: 0.75-0.80 # Yield formula: kWh =
System_kwp x Irradiance x PR x Days_in_month
```

SECTION C — CHALLENGE 1: CHATBOT ARCHITECTURE & TIPS

C1. Architecture Overview

The recommended architecture is a stateless webhook server that receives messages from Telegram or WhatsApp, routes them through a RAG pipeline, and returns responses. The core components are:

- **Bot Gateway:** Telegram Bot API (python-telegram-bot) or WhatsApp Cloud API / Twilio Sandbox. Receives user messages as webhooks.
- **Session Manager:** Redis or in-memory dict keyed by chat_id. Stores last 5 message pairs for conversation context. TTL: 30 min.
- **Intent Router:** Simple keyword classifier or LLM-based intent detection. Routes to: FAQ, Lead Capture, Escalation, or Fallback.
- **RAG Pipeline:** Query embedded against vector store (ChromaDB local or Pinecone free tier). Top 3 chunks + query sent to LLM.
- **LLM Layer:** OpenAI GPT-4o-mini (cheap, fast) or Claude Haiku (via Anthropic API). System prompt enforces SuperCharge persona.
- **Lead Capture:** Triggered when intent = lead. Collects name, email, enquiry type via multi-turn conversation. Writes to Google Sheet via gspread.
- **Escalation:** Triggered by: low confidence score, explicit 'speak to human', or 3 failed intents. Sends alert to Slack/email via webhook.

C2. Knowledge Base Construction

Your RAG knowledge base must be grounded in real SuperCharge SG content. Build it from:

- This Knowledge Base PDF (entire document — chunk by section)
- SuperCharge SG website content: supercharge.sg/electric-vehicle-charging/ and supercharge.sg/solar-installation/
- FAQs: compile from the 'Got Questions' section on the SuperCharge website
- Product specs: EV charger types (7kW AC, 22kW AC, 50kW DC), solar system sizes, pricing ranges
- Singapore-specific facts from Section B and G of this document
- Process guides: TR25 compliance steps, solar installation timeline, ECIS application process

■ Chunking strategy

Chunk by logical topic, not by fixed token count. Target 200–400 tokens per chunk. Include metadata (source, topic category) on each chunk. For FAQ content, keep Q and A together in one chunk. Use sentence-transformers (all-MiniLM-L6-v2) for embeddings — it's free, fast, and runs locally. Retrieve top 3 chunks and pass them to the LLM in the system context.

C3. Telegram Bot — Quick Start

```
# 1. Create bot: message @BotFather on Telegram, /newbot # 2. Install: pip install python-telegram-bot==20.x from
telegram import Update from telegram.ext import ApplicationBuilder, CommandHandler, MessageHandler, filters,
ContextTypes async def handle_message(update: Update, context: ContextTypes.DEFAULT_TYPE): user_msg =
update.message.text chat_id = update.effective_chat.id # 1. Get session history from Redis/dict # 2. Run intent
detection # 3. If FAQ intent: run RAG pipeline, get LLM response # 4. If lead intent: start lead capture flow # 5. If
escalation: fire webhook, reply with handoff message response = await rag_pipeline(user_msg, chat_id) await
update.message.reply_text(response) app = ApplicationBuilder().token('YOUR_BOT_TOKEN').build()
app.add_handler(MessageHandler(filters.TEXT & ~filters.COMMAND, handle_message)) app.run_polling()
```

C4. WhatsApp Business API Setup (+20 Bonus Points)

There are two ways to get a WhatsApp bot working. Option B is harder but earns the bonus:

Option	Setup Complexity	Approach	Bonus Eligible?
--------	------------------	----------	-----------------

Twilio WhatsApp Sandbox	Low — 15 min setup	Sign up at twilio.com → Messaging → WhatsApp Sandbox. Use ngrok for local webhook. Free sandbox number.	YES — earns +20 pts
Meta WhatsApp Cloud API	Medium — 1–2 hours	Create Meta Business account → Meta for Developers → WhatsApp product → get phone number. Webhook must be HTTPS.	YES — earns +20 pts
Telegram only	Lowest	python-telegram-bot, no business account needed.	NO — base points only

```
# Twilio WhatsApp Sandbox – webhook handler (FastAPI) from fastapi import FastAPI, Form from
twilio.twiml.messaging_response import MessagingResponse app = FastAPI() @app.post('/whatsapp') async def
whatsapp_webhook(Body: str = Form(), From: str = Form()): response_text = await rag_pipeline(Body, From) # your RAG
function resp = MessagingResponse() resp.message(response_text) return Response(content=str(resp),
media_type='application/xml') # Run locally: uvicorn main:app --port 8000 # Expose with: ngrok http 8000 # Set ngrok
URL as Twilio sandbox webhook: https://xxx.ngrok.io/whatsapp
```

C5. RAG Pipeline — Recommended Implementation

```
# Minimal RAG pipeline with ChromaDB (local, no API key needed) # pip install chromadb langchain sentence-transformers
openai import chromadb from langchain.text_splitter import RecursiveCharacterTextSplitter from sentence_transformers
import SentenceTransformer from openai import OpenAI # 1. Build knowledge base (run once) embedder =
SentenceTransformer('all-MiniLM-L6-v2') client = chromadb.PersistentClient(path='./chroma_db') collection =
client.get_or_create_collection('supercharge_kb') # Add chunks from your KB documents splitter =
RecursiveCharacterTextSplitter(chunk_size=400, chunk_overlap=50) # chunks = splitter.split_text(your_kb_text) #
embeddings = embedder.encode(chunks).tolist() # collection.add(documents=chunks, embeddings=embeddings, ids=[str(i)
for i in range(len(chunks))]) # 2. Query at runtime def rag_pipeline(user_query: str, history: list) -> str: query_emb
= embedder.encode([user_query]).tolist() results = collection.query(query_embeddings=query_emb, n_results=3) context =
'\n\n'.join(results['documents'][0]) oai = OpenAI() messages = [ {'role': 'system', 'content': f'You are a helpful
SuperCharge SG assistant. ' f'Answer ONLY from this context:\n{context}\nIf unsure, say so honestly.'}, *history[-4:],
# last 2 turns {'role': 'user', 'content': user_query} ] resp = oai.chat.completions.create(model='gpt-4o-mini',
messages=messages) return resp.choices[0].message.content
```

C6. Deployment — Free Hosting Options

Platform	Free Tier	Best For	Notes
Railway.app	500 hrs/mo free	FastAPI or Node.js bot	Easiest FastAPI deploy. Add env vars for tokens.
Render.com	750 hrs/mo free	FastAPI with background tasks	Supports webhooks well. Sleep on inactivity.
Fly.io	3 shared VMs free	Docker container	Good for persistent ChromaDB volume.
Vercel	Unlimited free	Next.js or serverless Python	Serverless only — no persistent file system.
HuggingFace Spaces	Free with GPU option	Streamlit or Gradio demos	Great for dashboard demos, less for bots.
ngrok (local)	Free tunnel	Development/testing only	Not for submission — evaluator can't access reliably.

SECTION D — CHALLENGE 2: SMART ENERGY DASHBOARD — ARCHITECTURE & TIPS

D1. System Architecture

The Smart Energy Dashboard has 5 architectural layers. Design them independently so each can be tested:

- **Data Simulation Layer:** Python scripts generating realistic solar and EV telemetry. Pushes to a time-series database every 30 seconds. Can inject faults on demand.
- **Backend API:** FastAPI exposing REST endpoints: /sites, /solar/timeseries, /ev/sessions, /anomalies, /reports. JWT auth for multi-tenancy.
- **Real-time Transport:** WebSockets (FastAPI WebSocket or Socket.IO) for live dashboard updates. Alternatively, Server-Sent Events (SSE) for simpler setup.
- **Frontend Dashboard:** React with Recharts or Chart.js. Shows live gauges, time-series charts, anomaly alerts, site selector, ECIS tracker.
- **AI & Reporting Layer:** Weekly digest via LLM API (scheduled). Monthly PDF via ReportLab. NASA POWER API for irradiance baseline.

D2. Data Model — Key Tables

```
-- Multi-tenant schema (PostgreSQL / Supabase)
CREATE TABLE organisations (id UUID PRIMARY KEY, name TEXT, created_at TIMESTAMPTZ);
CREATE TABLE sites (id UUID PRIMARY KEY, org_id UUID REFERENCES organisations(id), name TEXT, lat FLOAT, lng FLOAT, solar_kwp FLOAT, charger_count INT);
CREATE TABLE solar_readings (id BIGSERIAL, site_id UUID, ts TIMESTAMPTZ, power_kw FLOAT, energy_kwh FLOAT, irradiance FLOAT, temp_c FLOAT, expected_kw FLOAT, anomaly_flag BOOL, anomaly_severity TEXT);
CREATE TABLE ev_sessions (id BIGSERIAL, site_id UUID, charger_id TEXT, start_ts TIMESTAMPTZ, end_ts TIMESTAMPTZ, energy_kwh FLOAT, revenue_sgd FLOAT, status TEXT);
-- status: Available/Charging/Faulted
-- Row-level security: WHERE site_id IN (SELECT id FROM sites WHERE org_id = current_org_id())
```

D3. Anomaly Detection — Model Comparison

Model	Best For	Pros	Cons	Recommended ?
Isolation Forest	Point anomaly detection in multi-feature tabular data	scikit-learn, no training data needed, fast	No temporal awareness, needs feature engineering	YES — start here
Prophet (Facebook)	Seasonal time-series forecasting + anomaly flagging	Handles seasonality, interpretable, easy API	Slower, needs historical data (months)	YES — for production
LSTM Autoencoder	Sequence-level anomaly in temporal data	Captures patterns over time, powerful	Needs training data, GPU helpful, complex	Advanced only
Z-score / IQR	Simple threshold-based outlier detection	Trivial to implement, explainable	Too simple — misses subtle degradation	Baseline only

```
# Isolation Forest anomaly detection - minimal working example
# pip install scikit-learn numpy from sklearn.ensemble
import IsolationForest
import numpy as np
# Features: actual_kw, expected_kw, performance_ratio, temp_c, irradiance
def train_anomaly_model(historical_df):
    X = historical_df[['performance_ratio', 'actual_vs_expected_pct', 'temp_c']].values
    model = IsolationForest(contamination=0.05, random_state=42) # 5% expected anomaly rate
    model.fit(X)
    return model
def score_reading(model, reading):
    X = np.array([reading['performance_ratio'], reading['actual_vs_expected_pct'], reading['temp_c']])
    score = model.decision_function(X)[0] # negative = more anomalous
    is_anomaly = model.predict(X)[0] == -1
    severity = 'CRITICAL' if score < -0.3 else 'WARNING' if score < -0.1 else 'OK'
    return {'anomaly': is_anomaly, 'severity': severity, 'score': score}
```

D4. Multi-Tenant Authentication Pattern

```
# FastAPI JWT multi-tenant auth # pip install python-jose[cryptography] passlib[bcrypt] from fastapi import Depends,
HTTPException from jose import JWTError, jwt SECRET_KEY = 'your-secret' ALGORITHM = 'HS256' def create_token(org_id:
str, user_id: str) -> str: payload = {'sub': user_id, 'org_id': org_id} return jwt.encode(payload, SECRET_KEY,
algorithm=ALGORITHM) async def get_current_org(token: str = Depends(oauth2_scheme)) -> str: try: payload =
jwt.decode(token, SECRET_KEY, algorithms=[ALGORITHM]) return payload['org_id'] # use this to scope ALL DB queries
except JWTError: raise HTTPException(status_code=401) # In every endpoint: only query rows WHERE org_id = current_org
@app.get('/solar/readings') async def get_readings(org_id: str = Depends(get_current_org), db=Depends(get_db)): return
db.query(SolarReading).join(Site).filter(Site.org_id == org_id).all()
```

D5. AI Weekly Digest Generation

```
# Weekly digest prompt pattern def generate_weekly_digest(site_data: dict) -> str: prompt = f""" You are the energy
reporting assistant for SuperCharge SG. Write a professional weekly energy digest for client:
{site_data['client_name']} Site: {site_data['site_name']} | Week ending: {site_data['week_end']} DATA: - Solar
generated: {site_data['solar_kwh']:.1f} kWh (expected: {site_data['expected_kwh']:.1f} kWh) - EV sessions:
{site_data['ev_sessions']} sessions, {site_data['ev_kwh']:.1f} kWh delivered - ECIS credits earned: SGD
{site_data['ecis_credits']:.2f} - Anomalies detected: {site_data['anomaly_count']} ({site_data['anomaly_details']}) -
CO2 saved: {site_data['co2_kg']:.1f} kg Write 3 short paragraphs: (1) summary performance, (2) anomalies and actions,
(3) savings highlight. Keep it under 150 words. Plain English. No jargon. """ # Call your LLM here (OpenAI, Claude,
etc.) return call_llm(prompt)
```

SECTION E — CHALLENGE 3: AGV BRAIN — DESIGN REFERENCE

E1. System Overview & Design Philosophy

The AGV Brain is a universal autonomy module that intercepts and overrides the PWM signal lines between an industrial RC receiver (TADA B1900 class) and a grass cutter's drive systems. The key design principle is **non-destructive intercept**: the RC receiver must continue to function normally in MANUAL mode — the AGV Brain sits transparently in the signal path and only takes over when switched to AUTONOMOUS mode. If the AGV Brain loses power or crashes, the system must automatically return to MANUAL mode (fail-safe).

E2. TADA B1900 Full Signal Reference

The TADA B1900 (as visible on the product label in the provided photo) has the following terminals and signal characteristics:

Terminal	Signal Type	Typical Voltage / Pulse	AGV Brain Action	Safety Class
Engine Start (+/-)	Dry contact relay	N.O. contact, closes to start engine	GPIO + relay: read in MANUAL, replicate in AUTO	CRITICAL
Engine Stop (+/-)	Dry contact relay	N.O. contact, closes to stop engine	ALWAYS honour — never override E-stop signal	SAFETY-CRITICAL
DC Electric Push Rod 1 (+/-)	PWM motor drive	24V, PWM 500Hz–2kHz, 0–100% duty	PWM capture → motor driver H-bridge in AUTO	HIGH
DC Electric Push Rod 2 (+/-)	PWM motor drive	24V, PWM 500Hz–2kHz, 0–100% duty	Same as Push Rod 1	MEDIUM
DC 24V Input (+/-)	Power supply	24V DC, 5–15A system draw	Buck converter to 5V (RPI) and 12V (sensors)	HIGH
Signal Output 1 (JST 6-pin)	RC PWM servo signal	5V logic, 50Hz, 1000–2000µs pulse width	PWM capture with oscilloscope or logic analyser first	HIGH
Signal Output 2 (JST 6-pin)	RC PWM servo signal	5V logic, 50Hz, 1000–2000µs pulse width	Same as Signal Output 1	MEDIUM
Throttle Signal Output (+/-)	PWM or 0–5V analog	Verify with multimeter on your specific cutter	ADC or PWM capture, replicate in AUTO	HIGH
Damper Signal Output (+/-)	PWM or 0–5V analog	Verify on hardware	Intercept for blade engagement control	MEDIUM
ANT (SMA)	RF antenna	N/A — RF only	Do NOT modify — pass through untouched	N/A
ID Pair button	Momentary push	Grounds pair pin momentarily	Ignore in normal operation	LOW

E3. Compute Module Comparison

Module	CPU	RAM	PWM I/O	AI/ML	Cost (SGD)	Recommendation
RPi 4B + STM32F4	ARM Cortex-A72 1.8GHz + STM32 168MHz	4–8GB + 192KB	STM32 handles 16 PWM channels in real-time	ROS2 Nav2 on RPi	~\$80 + \$25	Best for this project — RPi handles nav, STM32 handles PWM timing
RPi CM4	ARM Cortex-A72 1.5GHz (embedded)	4–8GB	Needs HAT for PWM — recommend STM32 co-processor	Same as RPi 4B	~\$85 + \$25	Cleaner form factor. Same split-responsibility approach.
Jetson Nano Orin	6-core Arm Cortex-A78AE	8GB LPDDR5	Needs STM32 for PWM	Runs YOLOv8 for obstacle detect on-device	~\$250	Overkill unless you implement camera-based obstacle detection
Arduino Mega only	ATmega2560 16MHz	8KB SRAM	54 digital I/O, hardware PWM	No — too weak for navigation	~\$20	Insufficient for navigation stack — use as PWM co-processor only

E4. Navigation Stack — ROS2 Nav2 vs ArduPilot Rover

Criterion	ROS2 Nav2	ArduPilot Rover
Target use case	Indoor/outdoor mobile robots, well-mapped environments	Outdoor ground vehicles, GPS-reliant navigation
GPS support	Requires additional configuration, better for SLAM	Native RTK-GPS support, designed for GPS-first
Sensor integration	Excellent — LiDAR, camera, IMU all supported natively	Good — GPS, compass, sonar. Less flexible for custom sensors
Learning curve	High — requires ROS2 knowledge	Medium — Mission Planner GUI makes waypoint setup easy
Obstacle avoidance	Excellent — costmap-based replanning	Basic — sonar/lidar proximity only
Grass cutting recommendation	Better if adding camera-based detection or SLAM mapping	Better for simple GPS waypoint mowing with geofence
Community support	Large robotics community	Large UAV/rover community, good documentation

■ AGV design tip

For a plug-and-play grass cutter brain, ArduPilot Rover is likely your better choice. It handles GPS waypoints, geofencing, and mission planning natively. You can use Mission Planner on a laptop for initial setup and then hand off to the RPi for autonomous execution. ROS2 Nav2 is better if you plan to add a camera for obstacle detection — which is a good stretch goal.

E5. PWM Intercept Circuit Design

The safest PWM intercept approach uses an analog multiplexer (MUX) IC. In MANUAL mode, the MUX connects the RC receiver output directly to the grass cutter actuator. In AUTONOMOUS mode, the MUX switches to the STM32's PWM output channel. A hardware switch or GPIO signal controls the MUX selection pin.

```
# Recommended components for PWM intercept # 74HC4053 – Triple 2-channel analog MUX/DEMUX (5V logic compatible) #
CD74HC4051 – 8-channel MUX if more channels needed # Pinout concept: # Pin A (RC Receiver PWM output) ----| # |---
74HC4053 MUX ----> Actuator (ESC/servo) # Pin B (STM32 PWM output) ----| ^ # | # SELECT pin (GPIO from RPi) # LOW =
MANUAL (pass RC through) # HIGH = AUTO (pass STM32 through) # STM32 reads RC PWM in MANUAL mode (for monitoring &
learning) # In AUTO mode, STM32 generates PWM from navigation commands: # 1500µs = neutral/stop, 1000µs = full
reverse, 2000µs = full forward # For servo: 1000-2000µs range, 50Hz frequency # CRITICAL safety: hardware pull-down on
SELECT pin # If RPi crashes, SELECT defaults to LOW → MANUAL mode restored
```

E6. Sample BOM — Starting Template

Use this as your starting BOM. You must complete all rows with real supplier links and verified SGD pricing. Target total under SGD 900.

#	Component	Spec	Supplier (SG)	Unit SGD	Qty	Total SGD	Role
1	Raspberry Pi 4B (4GB)	ARM Cortex-A72, 4GB RAM, Wi-Fi	CytronEdu / Lazada	~\$85	1	\$85	Main compute — ROS2 / ArduPilot
2	STM32F4 Discovery Board	STM32F407, 168MHz, PWM x16	Mouser / AliExpress	~\$28	1	\$28	Real-time PWM capture + output
3	u-blox F9P RTK GPS	RTK <2cm accuracy, UART/I2C	AliExpress	~\$180	1	\$180	Precise positioning for waypoints
4	RPLidar A1M8	360deg 2D LiDAR, 12m range	AliExpress / Cytron	~\$120	1	\$120	Obstacle detection
5	MPU-9250 IMU	9-axis IMU, SPI/I2C	AliExpress	~\$8	1	\$8	Heading, tilt detection
6	74HC4053 MUX (x5)	Triple 2-ch analog MUX, 5V	RS Components / Mouser	~\$2	5	\$10	PWM signal intercept switching
7	Meanwell DRS-60-24	DIN rail 24V→5V/12V DC-DC	RS Components	~\$45	1	\$45	Compute power from 24V rail
8	SIM7600G 4G Module	4G LTE + GPS, UART	AliExpress	~\$55	1	\$55	Remote monitoring + telemetry
9	HC-SR04 Ultrasonic (x4)	2–400cm range, 5V	Lazada / AliExpress	~\$3	4	\$12	Front/rear bump detection
10	Tilt Switch SW-520D (x2)	Vibration/tilt sensor	AliExpress	~\$1	2	\$2	Blade stop on rollover
11	DIN Rail Enclosure	200x120x75mm, IP54	RS Components	~\$35	1	\$35	Weatherproof housing

12	Emergency stop button	Latching, NC contact, 22mm	RS Components	~\$18	1	\$18	Physical E-stop
13	JST connectors + wiring	JST-XH 2-6 pin, 22AWG wire	AliExpress	~\$15	1	\$15	Wiring harness
14	Capacitor bank (supercap)	2.7V 100F x4 (series = 5.4V)	AliExpress	~\$20	1	\$20	Graceful shutdown power buffer
— TOTAL						\$633	Under SGD 900 target ✓

SECTION F — EVALUATION CRITERIA & HOW TO EARN FULL MARKS

F1. Challenge 1 — Chatbot (35 pts + 20 WhatsApp bonus)

Live Deployment — 10 pts — 10/10

- Bot responds in under 5 seconds. No 502 errors. Available throughout the evaluation window. Tested on Telegram (base) or WhatsApp (bonus). Evaluator will run 10+ test messages.

RAG Accuracy — 10 pts — 10/10

- Evaluator asks: 'What does a 5kWp solar system cost?', 'What is TR25?', 'Can I charge my EV overnight?', 'What is ECIS?', 'Do you cover HDB installations?' — all answers must be grounded in the KB. Hallucinated facts (wrong prices, invented features) = immediate deduction.

Intent Coverage — 5 pts — 5/5

- Successfully handles: FAQ, Lead Capture, Price Enquiry, Fault/Maintenance Report, Human Agent Request, Greeting. 1 point per intent handled correctly.

Lead Capture — 5 pts — 5/5

- During test, evaluator says 'I'm interested in solar panels'. Bot must collect: name, email, enquiry type. Data must appear in Google Sheet within 10 seconds. Confirmation sent to user.

Escalation — 5 pts — 5/5

- Evaluator says 'I need to speak to a person'. Bot responds with handoff message AND a Slack/email alert is sent with conversation summary. Evaluator will verify alert received.

WhatsApp BONUS — +20 pts — +20

- Bot deployed and functional on WhatsApp (Twilio sandbox or Cloud API). All 5 base criteria tested on WhatsApp.

F2. Challenge 2 — Dashboard (35 pts)

Live Dashboard + Real-time Data — 8 pts — 8/8

- Evaluator opens URL in browser. Dashboard loads within 3 seconds. Charts update without page refresh. At least 3 sites visible. Solar and EV data present for each site.

ML Anomaly Detection — 9 pts — 9/9

- Evaluator will trigger a fault: set one site's simulated solar output to 40% below expected for 1 hour. The anomaly must be flagged with severity (Critical/Warning) within one data refresh cycle. Anomaly visible on dashboard. Model methodology explained in README.

Multi-Tenant Isolation — 7 pts — 7/7

- Login as Client A — see only Client A sites. Login as Client B — see only Client B sites. Attempt to access Client B endpoint with Client A JWT — must return 403. Evaluator will test all three scenarios.

AI Weekly Digest — 5 pts — 5/5

- Forward a sample generated digest email to recruiter. Must be: accurate (numbers match dashboard), non-generic (mentions specific site names and anomalies), client-ready (professional tone, under 200 words).

Monthly PDF Report — 4 pts — 4/4

- Download a sample PDF from the dashboard. Must include: client name, site summary table, savings chart, anomaly log, ECIS credits. Branded with SuperCharge SG visual identity.

ECIS Accuracy — 2 pts — 2/2

- ECIS credit calculation = $\text{exported_kWh} \times 0.218$ (or current rate). Formula documented in README. Evaluator will manually verify one site's credit against the raw data.

F3. Challenge 3 — AGV Brain (30 pts)

System Architecture Clarity — 6 pts — 6/6

- Block diagram shows full signal chain: RC Transmitter → TADA B1900 → AGV Brain → actuators. All 11 TADA B1900 terminals addressed. Signal flow arrows clear. MANUAL/AUTO paths distinct.

PWM Intercept Design — 6 pts — 6/6

- Non-destructive intercept method specified. MUX IC or relay-based approach with component names. Failsafe-to-MANUAL on power loss described. Engine Stop signal always honoured.

Navigation Stack — 5 pts — 5/5

- Stack choice (ROS2 Nav2 or ArduPilot) justified with specific reasons for grass cutting use case. Sensor selection (GPS, LiDAR, IMU) matched to the navigation approach.

Safety Interlocks — 5 pts — 5/5

- All five addressed: tilt stop, bump/E-stop, geofence boundary, dead-man switch logic, failsafe-to-manual. Each with a specific component or software approach.

BOM — 5 pts — 5/5

- All components listed with real specs, real supplier names, real SGD unit costs (sourced, not estimated). Total under SGD 900. Role of each component explained.

Integration Guide — 3 pts — 3/3

- Wiring diagram maps each TADA B1900 terminal to a specific AGV Brain input/output. Step-by-step instructions clear enough for a technician without additional questions.

SECTION G — SINGAPORE MARKET & DOMAIN QUICK REFERENCE

G1. Singapore EV Market Facts (for Chatbot KB)

- Singapore targets 60,000 EV charging points by 2030 (LTA national goal)
- As of 2024, approximately 8,000–10,000 EVs registered in Singapore (growing ~30% YoY)
- LTA's ECCG (EV Common Charger Grant) provides subsidies for condo and carpark charger installation
- HDB carparks are getting EV chargers through the HDB-LTA ECCG pilot programme
- Average Singapore driver travels 50–60 km/day — a 7kW AC charger fully charges most EVs overnight
- Singapore electricity tariff (2024): approximately SGD 0.32–0.36/kWh for residential users
- Peak demand periods: 7–10 AM and 6–10 PM — smart charging should avoid these times
- Type 2 (Mennekes) is the standard AC connector. CCS2 for DC fast charging. CHAdeMO being phased out.
- Tesla uses its own connector but sells CCS2 adapters for Singapore market
- EV charging cost via public chargers: ~SGD 0.40–0.65/kWh depending on operator and speed
- An EV with a 60kWh battery costs approximately SGD 19–22 to fully charge at home
- Compared to petrol: typical ICE car costs SGD 12–18/100km. EV costs SGD 3–5/100km on electricity.

G2. Singapore Solar Market Facts (for Chatbot KB)

- Singapore receives approximately 1,600–1,800 peak sun hours per year
- Average daily irradiance: 4.5–5.2 kWh/m²/day (varies by month — lowest in Nov/Dec)
- A 1kWp solar system generates approximately 100–120 kWh/month in Singapore
- Panel efficiency has improved to 22–23% for premium monocrystalline panels (2024)
- Typical landed home installs 3–8kWp. Commercial rooftops: 20–100kWp
- Payback period for residential solar: typically 8–12 years (after CPF/grants if applicable)
- SolarNova programme: Singapore's government initiative to install solar on HDB rooftops
- HDB rooftops are owned by HDB — residents cannot install solar on their own flat rooftop
- Landed homeowners need BCA approval for structural changes. SuperCharge handles this.
- EV + Solar bundle: solar self-consumption for EV charging maximises savings — avoids grid import cost
- Degradation rate of quality panels: approximately 0.5% per year (25-year warranty standard)
- Performance Ratio (PR) for Singapore installs: typically 0.75–0.80 (accounts for heat, inverter losses)

G3. Common Customer FAQs — Pre-written Answers for KB

Use these Q&A; pairs directly in your chatbot knowledge base. They are grounded in SuperCharge SG context.

Q: How long does EV charger installation take?

A: Most residential EV charger installations are completed in 1–2 days. Commercial or multi-unit sites take 3–5 days. Site assessment and approvals typically add 1–2 weeks before installation begins.

Q: Can I charge my EV overnight at home?

A: Yes. Our home chargers are designed for safe, unattended overnight charging with built-in protection circuits. Charging overnight during off-peak hours (11 PM–7 AM) also reduces your electricity cost.

Q: What is TR25 and do I need to worry about it?

A: TR25 is Singapore's technical standard for EV charging installations. SuperCharge handles all TR25 compliance — you don't need to worry about it. Our Licensed Electrical Workers certify every installation.

Q: Can I sell my solar energy back to the grid?

A: Yes, under Singapore's Enhanced Central Intermediary Scheme (ECIS), you can export excess solar energy to the grid and receive credits on your SP electricity bill. SuperCharge handles the SP Group application for you.

Q: How much does a solar system cost?

A: Residential solar systems (3–5kWp) typically cost SGD 8,000–14,000 installed. Commercial systems vary widely. We recommend scheduling a free site assessment for an accurate quote — every roof is different.

Q: How long will my solar panels last?

A: Quality solar panels are warranted for 25 years. Panels typically degrade less than 0.5% per year. After 25 years, most panels still operate at 80–85% of their original capacity.

Q: Do I need approval to install solar panels?

A: Yes, but SuperCharge handles all approvals — including SP Group grid connection approval, any BCA structural approvals for landed homes, and LTA or Town Council requirements. You just sit back and enjoy the savings.

Q: What is the difference between a 7kW and 22kW EV charger?

A: A 7kW single-phase charger is suitable for most homes — it fully charges most EVs in 6–10 hours overnight. A 22kW three-phase charger charges 3x faster (2–3 hours) but requires a 3-phase electrical supply, which most landed homes and commercial premises have.

G4. Useful API & Tool Links

Tool / API	URL	Used In	Notes
NASA POWER API	power.larc.nasa.gov/api	Ch.2 Dashboard	Solar irradiance data for Singapore. Free, no auth.
Twilio WhatsApp Sandbox	console.twilio.com	Ch.1 Chatbot	Free sandbox for WhatsApp bot testing.
Meta WhatsApp Cloud API	developers.facebook.com/docs/whatsapp	Ch.1 Chatbot	Free tier: 1,000 conversations/month.
OpenAI API	platform.openai.com	Ch.1+2	GPT-4o-mini: ~\$0.15/1M tokens. Very cheap for chatbot.
Anthropic Claude API	console.anthropic.com	Ch.1+2	Claude Haiku: even cheaper. Excellent for RAG.
ChromaDB	docs.trychroma.com	Ch.1 Chatbot	Local vector DB. No API key. Ideal for embedded RAG.
Pinecone	pinecone.io	Ch.1+2	Free tier: 1 index, 100k vectors. Good for cloud deploy.
Railway.app	railway.app	Ch.1+2	Easiest FastAPI cloud deploy. GitHub auto-deploy.
Supabase	supabase.com	Ch.2 Dashboard	Free PostgreSQL with REST API. Good for multi-tenant.
ArduPilot Mission Planner	ardupilot.org/planner	Ch.3 AGV	Free Windows/Mac GUI for ArduPilot Rover config.
draw.io	app.diagrams.net	Ch.3 AGV	Free diagram tool for block diagrams and wiring diagrams.
ReportLab	pypi.org/project/reportlab	Ch.2 Dashboard	Python PDF generation. Used in this Knowledge Base.